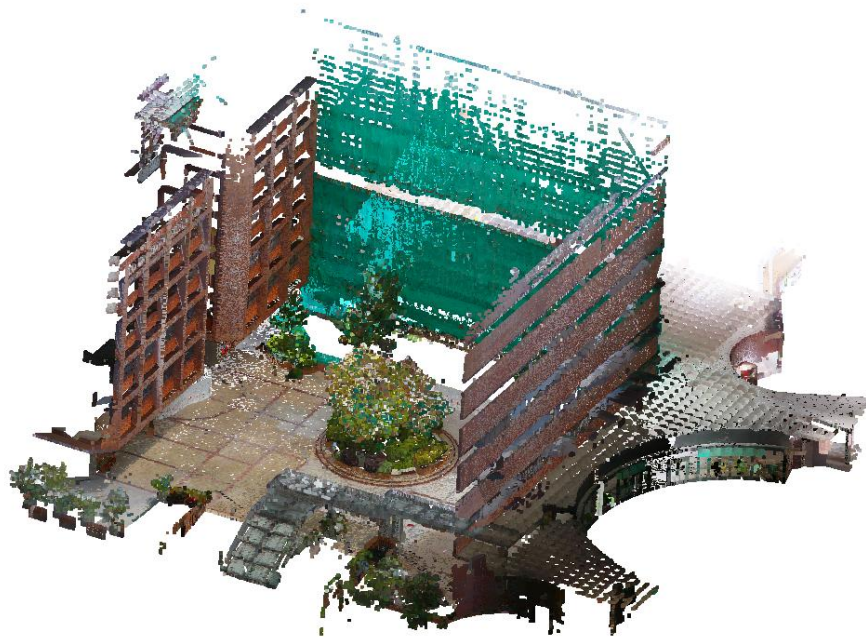


LSGI3262 – Laser Scanning

Group ■ - Group Report

Subject Lecturer: Prof. Yin Tiangang

Urban Features: 3D Point Cloud Acquisition and Semantic Segmentation of Block W Square



Students

Country	Share of GDP	Country	Share of GDP
United States	1.2%	United States	1.2%
Germany	0.8%	Germany	0.8%
France	0.6%	France	0.6%
Japan	0.4%	Japan	0.4%
United Kingdom	0.3%	United Kingdom	0.3%

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1. Introduction

Terrestrial Laser Scanning (TLS) has been considered a vital technology in surveying and geomatics today to obtain a high point cloud density for acquiring 3D point clouds in environments. This project is entitled "Urban Features: 3D Point Cloud Acquisition and Semantic Segmentation of Block W Square" where a survey of TLS using Leica BLK360 G2 was undertaken in Block W Square and 2/F covered plaza of TU Wing (TU plaza) of The Hong Kong Polytechnic University.

1.1 Primary Objectives

- To design and implement field surveys that will acquire a variety of urban features such as buildings, trees, circular planters, benches, ceilings, floors, pillars, scaffoldings, water-filled barriers, covered walkways, and recycling bins.
- To georeference the point clouds collected through RTK-GNSS control points.
- To preprocess raw data through removing noise, down sampling, and filling voids.
- To conduct semantic segmentation to classify major urban classes through automatic and manual segmentation.

1.2 Practical Applications

This example shows how TLS can be utilized in urban environments. Semantic segmentation in urban environments presents several difficulties because of occlusion caused by pillars, variable point density, and dynamic nature of some objects such as people. After segmentation, the classified point cloud can be used for different applications like urban planning, facility management, and creation of digital twins.

2. Methodology

2.1 Field Survey Planning

The scanning plan was done using Open3DHK and iB1000 basemaps to identify 15 positions for TLS scans. The distribution included:

- 6 scans in Block W Square
- 3 scans in the W-TU corridor
- 6 scans in the TU plaza

Additional scans were conducted on-demand basis for void-filling or increasing local point density.

After site reconnaissance, it was discovered that several locations had obstruction by pillars and other structures. To ensure proper registration through high overlap, the scanning stations in corners were placed nearby each other. The benches around the central circular flowerbed area could be under-sampled after conducting the first 13 scans. Thus, two more stations (S14 and S15) were planned to cover those benches.

Figure 2.1: 15 proposed TLS scanning locations on Open3DHK basemap

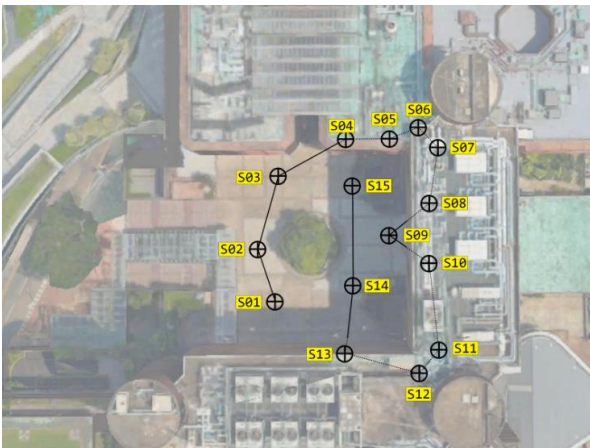


Figure 2.2: 15 proposed TLS scanning locations on iB1000 basemap

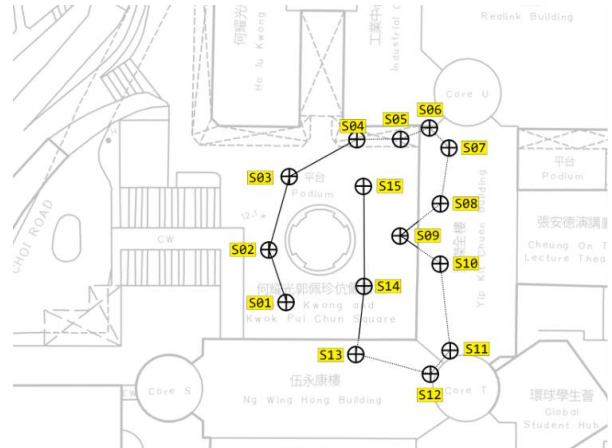
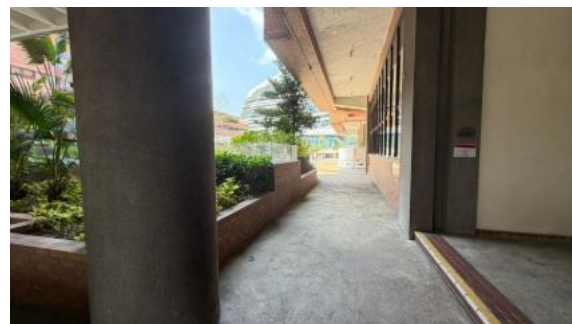


Figure 2.3: Photos showing intervisibility issues and additional scan positions (photos from proposed S07, S12, S14, S15 respectively, left to right, top to bottom)



2.2 TLS and RTK Field Surveys

The terrestrial laser scanner Leica BLK360 G2 was chosen due to its portability, fast data acquisition, and quality data suitable for urban areas.

In total, 15+ scans were made as per the initial plan. At each location, the scanning stations were leveled and oriented. High-resolution scans were conducted, and simultaneously RTK-GNSS surveying was done to obtain ground control points needed for georeferencing. The scanning targets were set at necessary locations to link the scans.

During the survey, it was noticed that some new areas got blocked by water-filled barriers and scaffoldings. The original scanning locations had to be changed in order to cover those places.

2.3 Raw Data Export and Georeferencing

The obtained raw scan data were exported from Leica Cyclone. The individual scans were linked using Cyclone automatic linking functionality. Georeferencing was done by incorporating the GNSS known points of physical features to transform the point cloud into the local Hong Kong 1980 grid coordinate system.

2.4 Denoising and Downsampling

The merged point cloud was uploaded to CloudCompare for preprocessing. First, down-sampling was done to reduce the amount of data for processing. Statistical outlier removal was conducted to filter out noisy points caused by movement of people. The preprocessing workflow in CloudCompare consisted of four steps:

1. Cloud sub sampling
 - Method: Spatial
 - Minimum space between points: 0.01m
 - Purpose: Reduce data density while maintaining feature representation
2. Statistical Outlier Removal (SOR)
 - Number of points to use for mean distance estimation: 20
 - Standard deviation multiplier threshold (nSigma): 1.00
 - Purpose: Remove outlier noise points
3. Filter by Value
 - Range kept: 500 to 65534

- Purpose: Remove extreme elevation outliers and invalid returns
4. Filter Noise (Final cleanup)
- Method: Radius (Sphere)
 - Sphere radius: 0.2m
 - Max error threshold: 0.020000 (absolute)
 - Purpose: Final cleanup of remaining scattered noise

Figure 2.4: CloudCompare – Cloud sub sampling

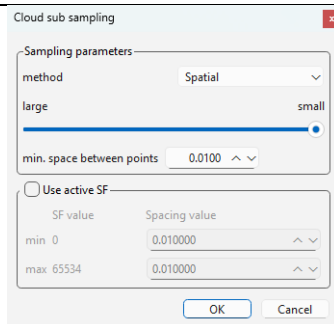


Figure 2.5: CloudCompare – Statistical Outlier Removal (SOR)

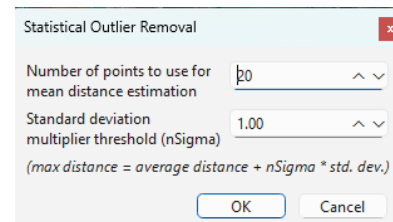


Figure 2.6: CloudCompare – Filter by Value

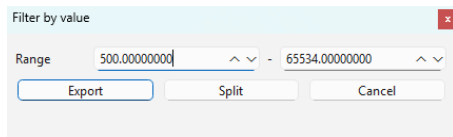


Figure 2.7: CloudCompare – Filter Noise

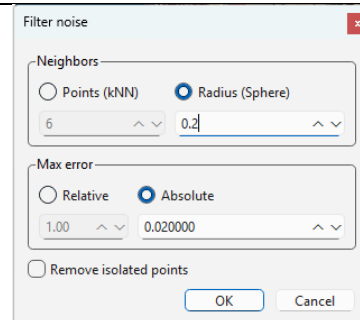
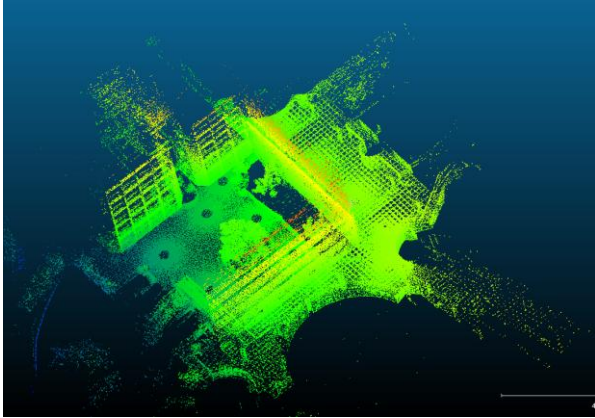
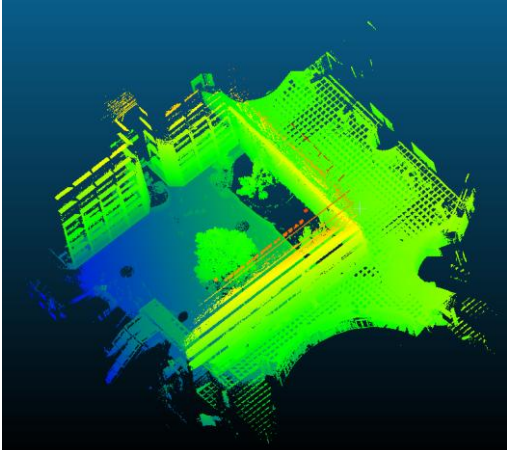


Figure 2.8: CloudCompare – Before preprocessing	Figure 2.9: CloudCompare – After cleaning and downsampling
	

2.4 Edge Trimming and Boundary Cleanup

After the denoising and downsampling is finished, it was imported the *las* file into Leica Cyclone 3DR. The manual edge trimming was performed to remove unwanted peripheral data:

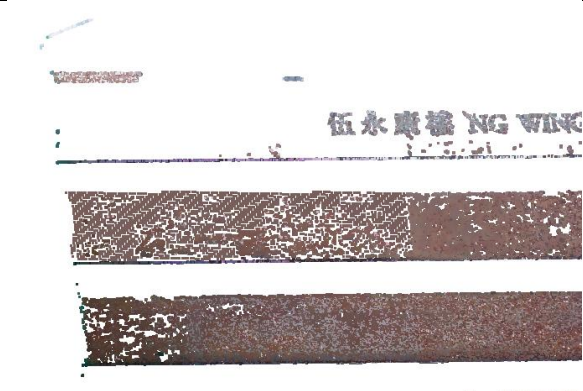
Edge Removal Process:

1. Used polygon selection tool to manually select edge regions
2. Deleted selected edge points to create clean boundaries

The edge trimming was done due to the reasons that the edge points had fewer point densities. There was incomplete feature coverage in the peripheral areas, and scan range limits introduced edge artifacts and noise. Additionally, it also eliminated excess data points outside the surveyed area. This cleanup step made sure that the following classification stage would focus on good quality points of cloud data within the surveyed area.

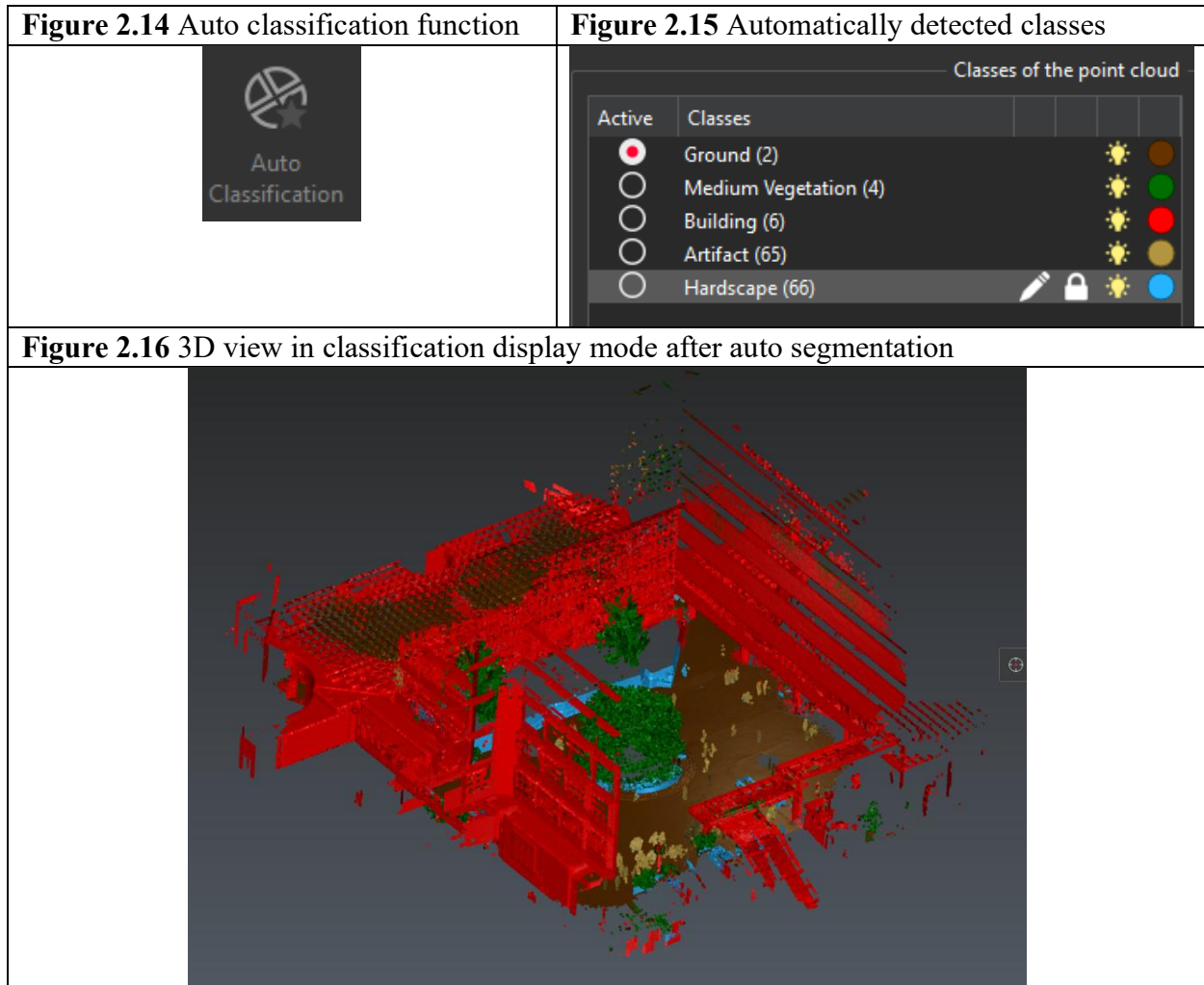
2.5 Void Filling

Minor voids caused by occlusions like areas below the tripod or obstruction were determined and filled using the Void Filling Function of 3DMas (point density = 0.05m). Additionally, manual void filling of voids was also done. Such actions ensure a more complete 3D model required for semantic segmentation.

Figure 2.10 A void caused by tripod obstruction	Figure 2.11 After filling the void in Figure 2.10
	
Figure 2.12 A building void caused by occlusion	Figure 2.13 After filling the void in Figure 2.12
	

2.6 Auto Segmentation with Cyclone

The Leica Cyclone built-in semantic segmentation tools were used initially to create the segmentation of the model. Machine learning classifiers of the software were used to assign preliminary labels to such major plane classes as ground, walls, and ceilings, and vegetation.



2.7 Segmentation with 3DMas

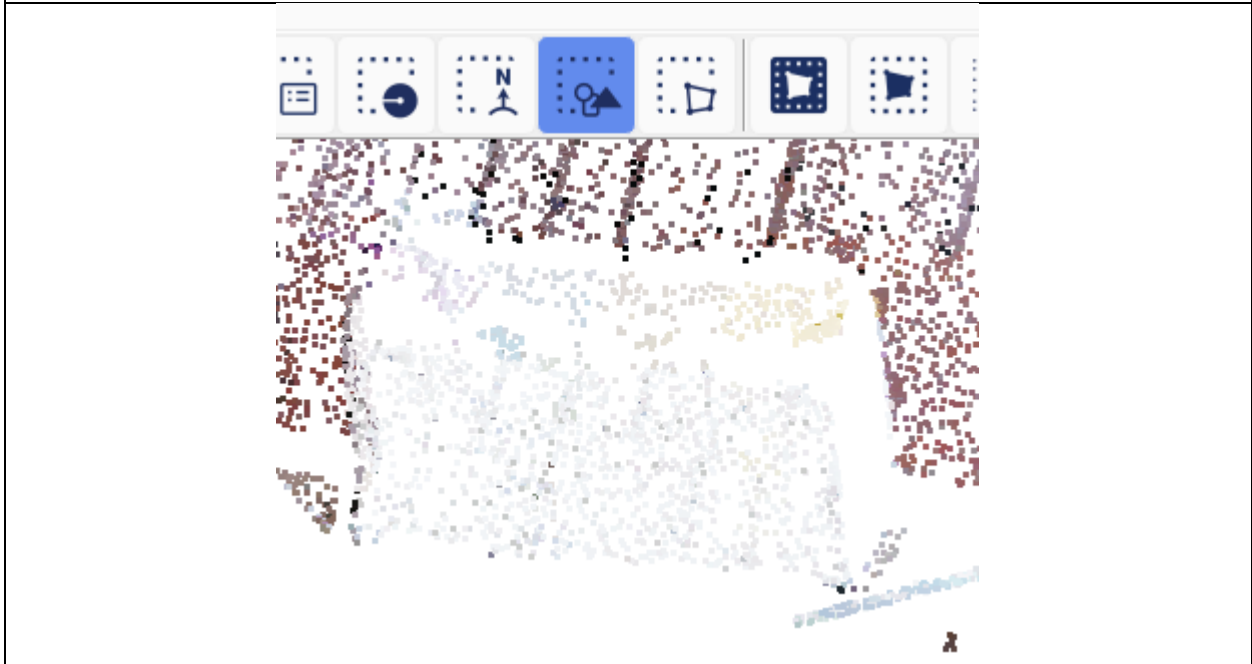
Afterwards, additional segmentation was done in 3DMas using both semi-supervised and manual methods.

2.7.1 Semi-supervised segmentation

Semi-supervised segmentation involves recognizing the boundaries of points with similar geometry and spectral characteristics in homogeneous regions where internal variation and noise are minimal. For this project, the one-click segment tool was used to separate points with similar characteristics in the selected target area.

The tool uses color intensity, elevation, and density to identify boundaries of distinct objects. With this tool, an object is separated by selecting one point within the area of interest.

Figure 2.17 One click segmentation to separate recycling bins from walls



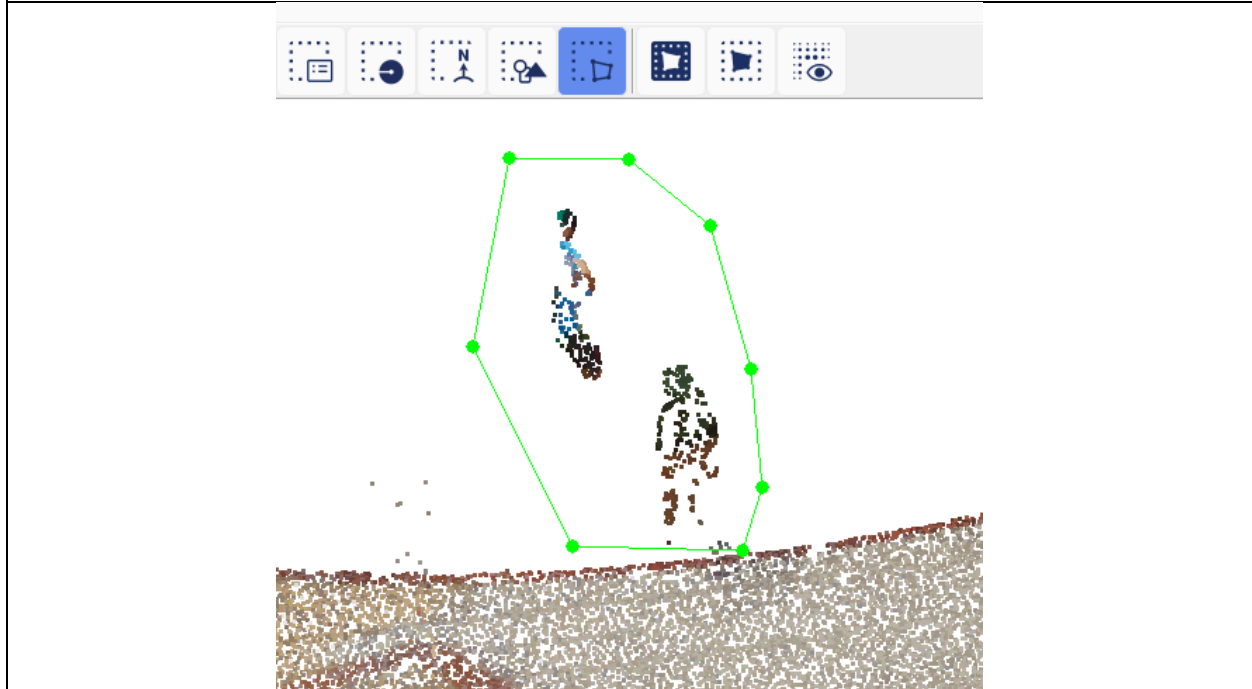
2.7.2 Manual segmentation

Polygon selection was also manually performed for points in challenging conditions in the area. It is mainly used for areas where points in which do not have the same intensity, shadowed areas, occluded areas, high noise areas, or low point density areas.

Some of the typical urban feature classes that require manual polygon selection include noise points within buildings, building-vegetation boundaries, sparse tree points that can be confused with ground cover points, small distinct objects like fences and poles.

With the polygon selection tool, the boundary points of an object can be segmented accurately. After the points of the polygon are selected, they merged to their respective categories. While automated one-click segmentation can save time during the initial phase of classification, manual refinement is necessary to ensure precise boundaries are achieved without mistakes.

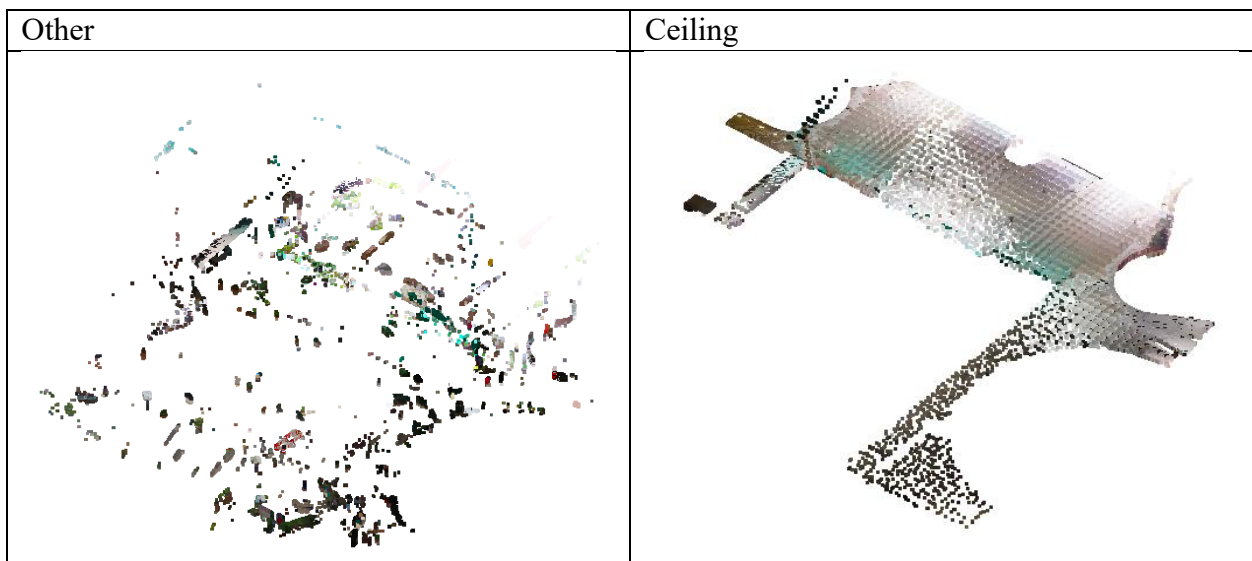
Figure 2.18 Using polygon selection tool to separate the human noises from the ground

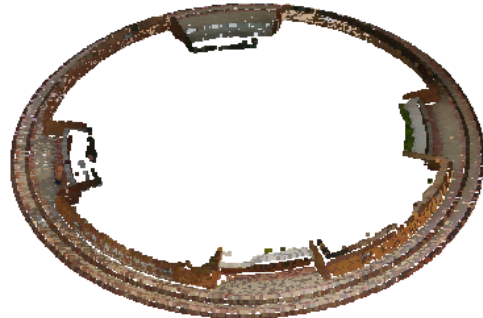
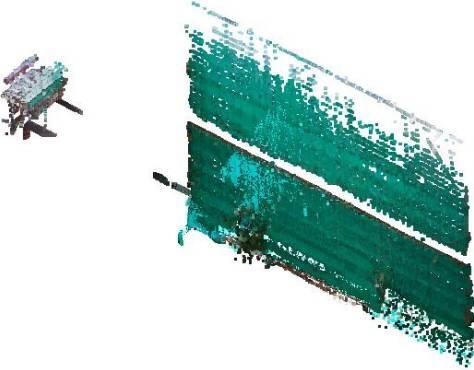
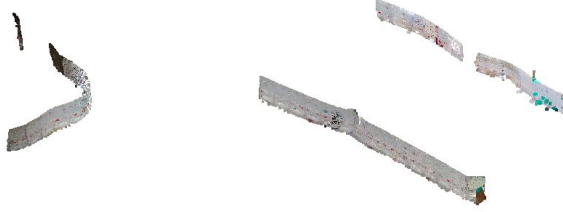


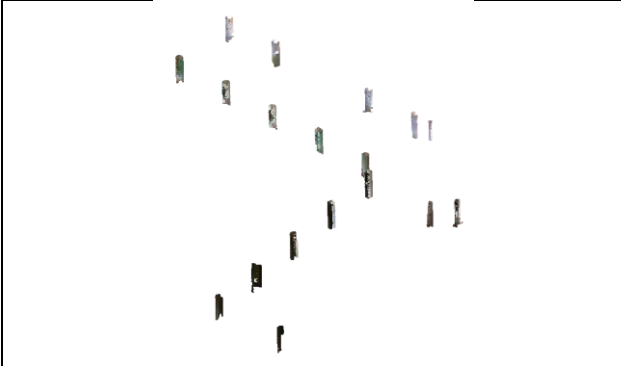
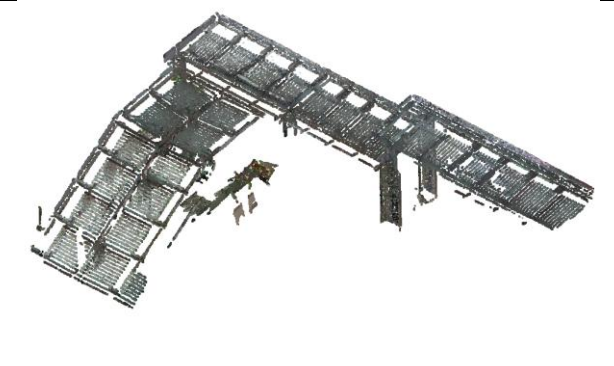
2.8 Visually Inspect results in RGB Mode

After segmenting into different classes, the results were inspected in RGB mode to ensure quality.

Below shows individual segmented classes in RGB mode:



Ground	Planter
	
Vegetation	Wall
	
Scaffolding	Barrier
	

Pillar	Covered Walkway
	

From the above, the segmentation has shown high quality: The ceilings and ground elevations were uniform; the vegetations were only in leafy and woody points; scaffoldings, walls, and barriers were vertical and have the same surface normal; the pillars looked uniform in discrete locations; covered walkway has shown continuity. The residual noise points are moved to “Others”, being the only class with high entropy. For the rest of the classes, they have shown minimal noises or leakages, which indicated the class boundaries were delineated precisely.

2.9 2D Map and DTM Production in ArcGIS Pro

2.9.1 2D Map

1. Create LAS Dataset

Create a .lasd file that references the original .las file. This feature allows ArcGIS Pro to read and display point cloud data directly in a 2D map without converting or importing the original points into a new format. It preserves all original point attributes (including classification) and supports visualization and filtering.

2. Symbolology of LAS Dataset Layer

The display method of the LAS dataset layers is changed from elevation-based color to classification-based color. Points are colored according to their classification (e.g., ground and vegetation), thus producing the colored 2D map view.

3. Manual Digitizing of Classification

Although point clouds offer precise and detailed spatial data, the color classification of those points based on categories (such as ground, vegetation, walls, and pillars) may not be entirely free from noise and overlapping points, which make it less clear. Therefore, manually digitizing the 2D maps from the classified point cloud is necessary.

This involves interpreting the classified point cloud layer visually and drawing polygons, lines, and points manually for every class, such as planter, vegetation, walls, scaffolds, barriers, pillars,

grass, trees, and covered walkways. The manual process effectively eliminates many overlapping points and provides a clearer 2D map.

2.9.2 DTM

After obtaining the classified point cloud, a digital terrain model (DTM) is generated using only ground points (Class 2).

1. Export ground points from 3DMas to ASCII (txt) format

In 3DMas, only the points belonging to Class 2 (ground) are exported and stored as ASCII text files (.txt), which store the XYZ coordinates and classification and the return number.

2. Import the ASCII file into ArcGIS Pro using XY Table to Point
3. Convert the point feature class to a raster

Use the Point to Raster tool to generate the DTM:

- Value field: elevation (Z)
- Cell assignment type: Mean
- Cell size: 0.27 m.

3. Results

3.1 Classification Display Mode

The fully labeled point cloud was displayed in classification mode, with each class shown in its assigned color. The result clearly separates semantic elements into distinct classes.

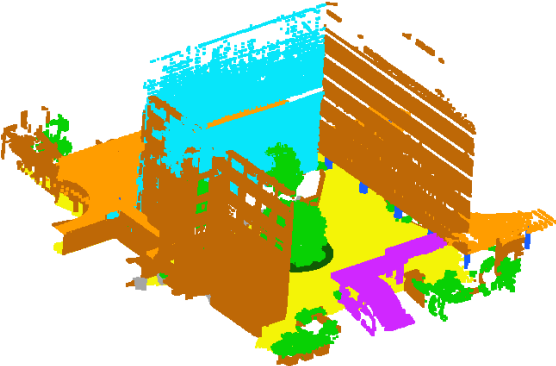
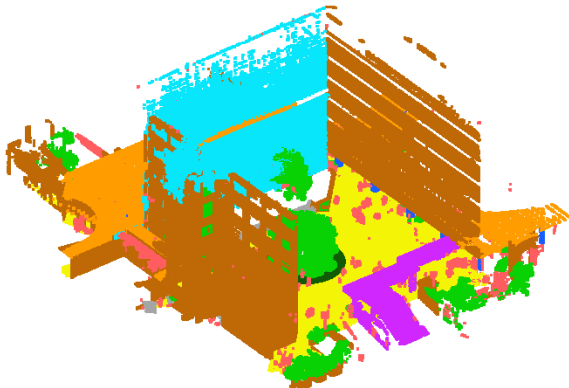
Figure 3.1 Classification display without "Other" class	Figure 3.2 Classification display with all classes
	

Figure 3.3 View from Core T facing Block Z, without Ceiling

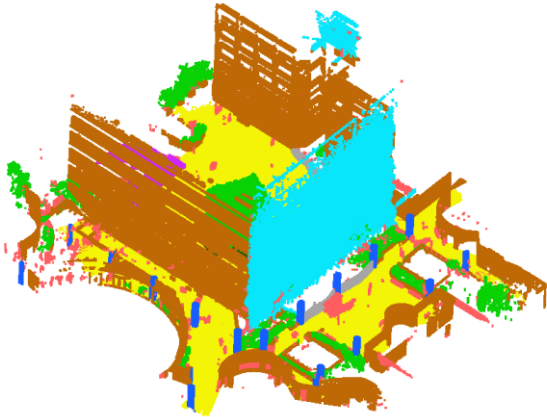


Figure 3.4 Top view without "Other" or "Ceiling"

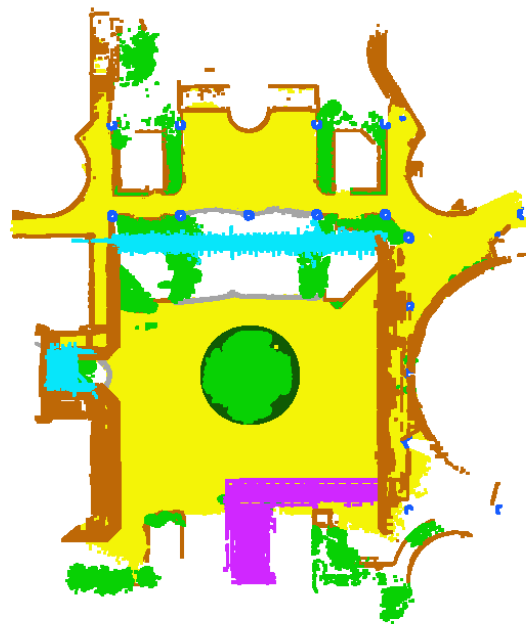


Figure 3.5 Legend of the 7 defined classes, their respective ID, map color, and point count.

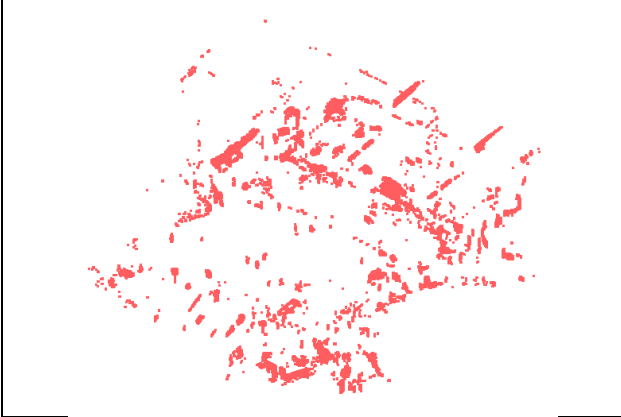
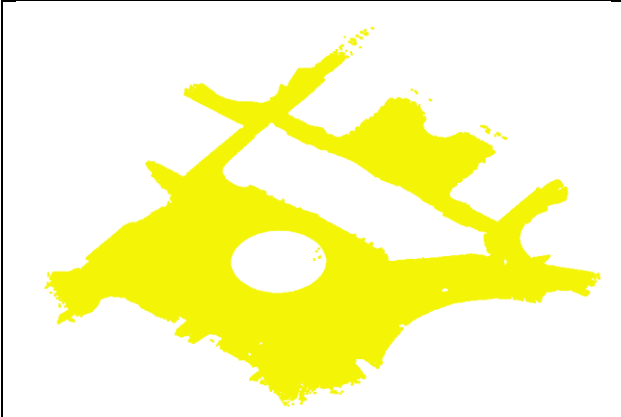
ID /	Class Name	Size	
0	Other	147,341	
1	Ceiling	841,876	
2	Ground	995,023	
3	Planter	38,095	
4	Vegetation	621,146	
5	Wall	1,888,053	
6	Scaffolding	909,597	
7	Barrier	59,654	
8	Pillar	95,468	
9	Covered Walkway	106,238	

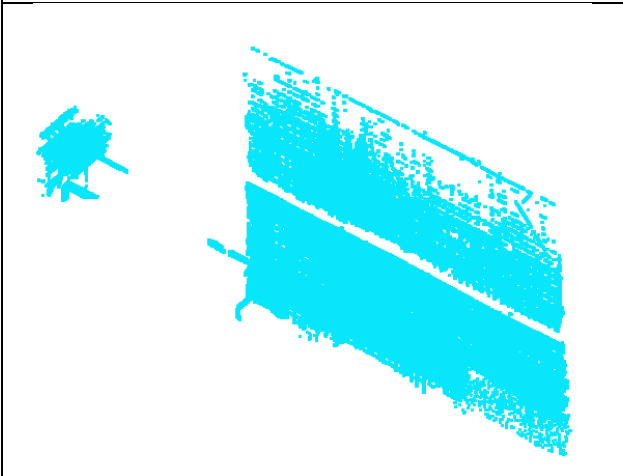
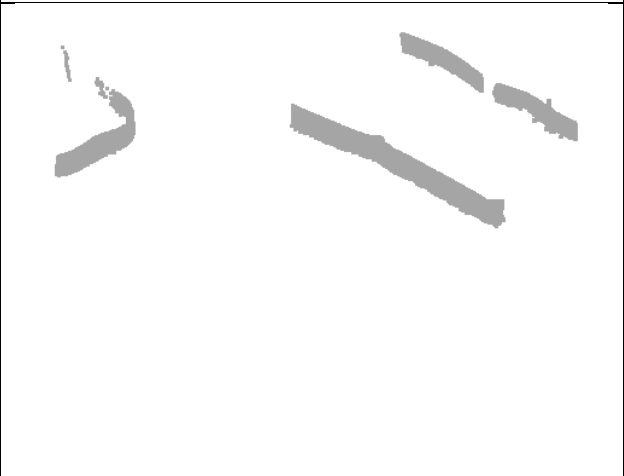
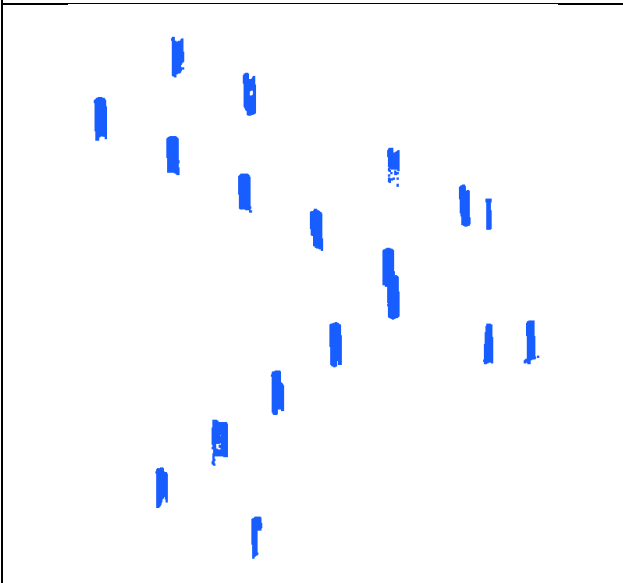
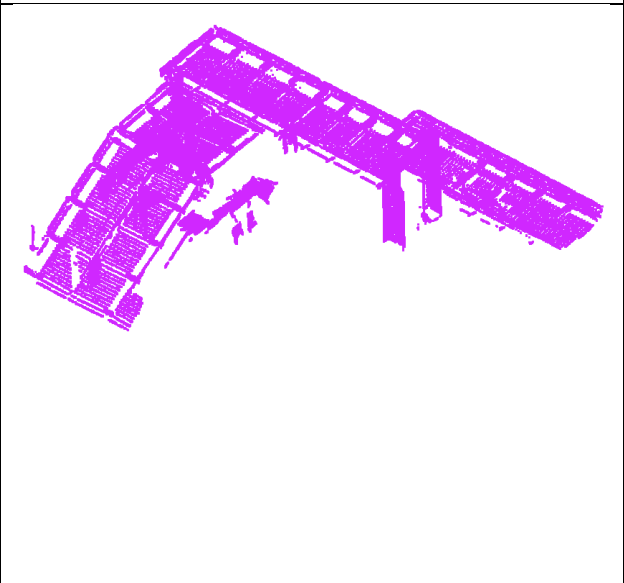
3.2 Classification Classes

The following semantic classes were defined for the urban environment:

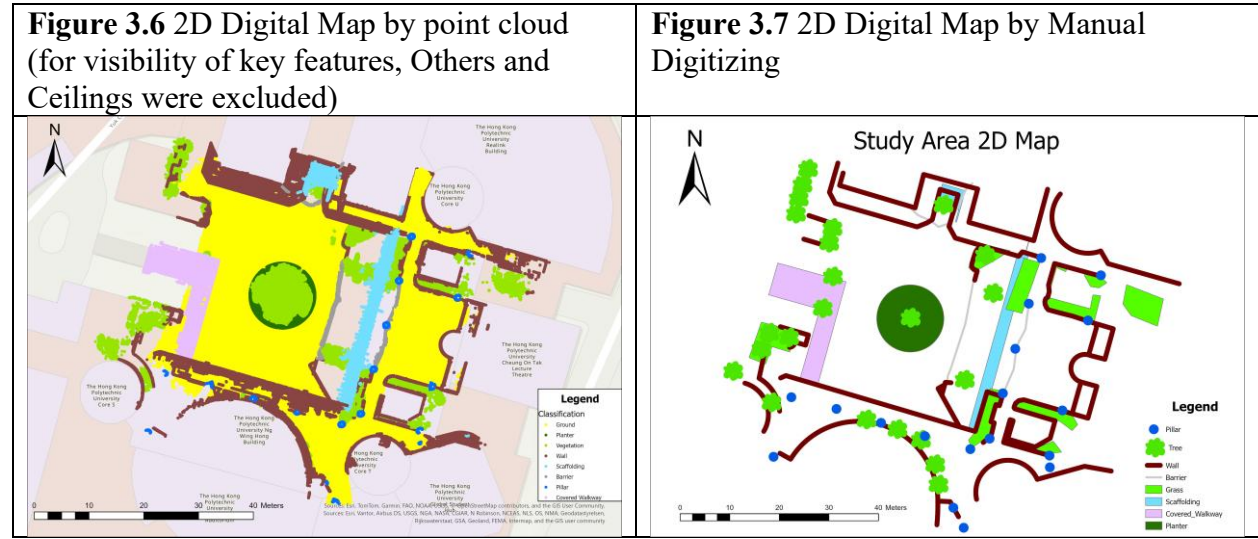
Class ID	Class Name	Color	Description
0	Other	Red	Miscellaneous objects and noise
1	Ceiling	Orange	Covered plaza ceilings
2	Ground	Yellow	Floor and pavement
3	Planter	Dark Green	Block W Square circular planter with benches
4	Vegetation	Green	Trees and shrubs
5	Wall	Brown	Buildings and glazed walls
6	Scaffolding	Cyan	Scaffolding bamboo structure and scaffold nets
7	Barrier	Gray	Water filled barriers
8	Pillar	Blue	Circular support columns of the buildings
9	Covered Walkway	Purple	Sheltered pedestrian walkway

3.3 Classification Display Mode for each Segmented Class

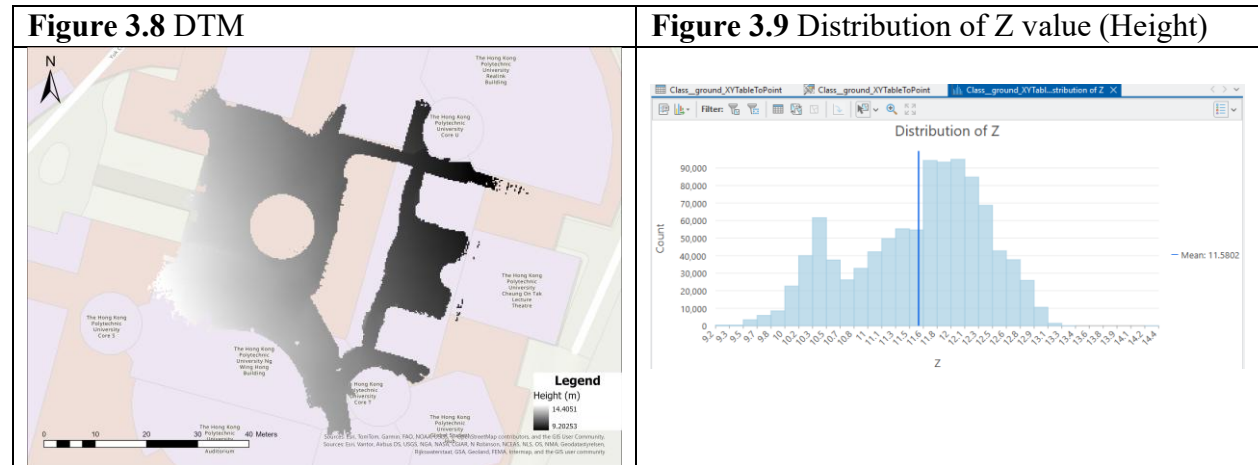
Other	Ceiling
	
Ground	Circular Planter
	

Vegetation	Wall
	
Scaffolding	Water-filled barriers
	
Pillars	Covered Walkway
	

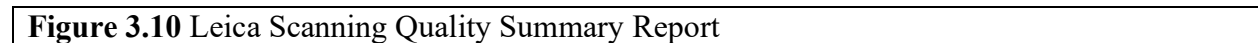
3.4 2D Digital Map

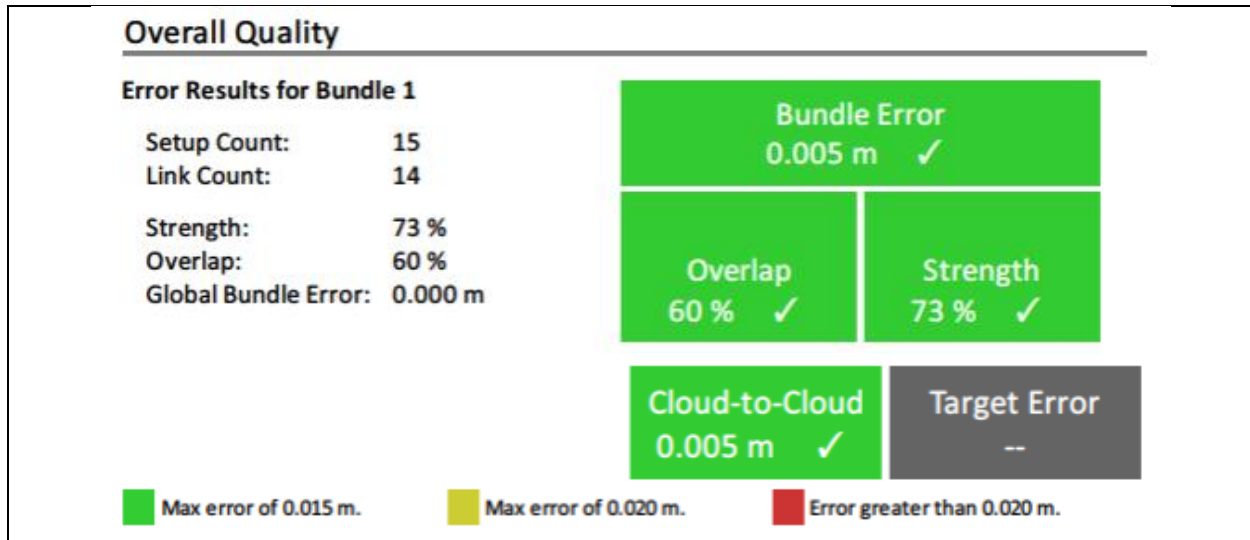


3.5 Digital Terrain Model



3.6 Leica Scanning Quality Summary





4. Analysis

4.1 Segmentation Performance and Quality

By combining automatic and manual processing, this project has achieved high-quality urban classes. Automatic segmentation removed major artifacts caused by pedestrians walking past the surveying setup. One-click classification (semi-supervised) performed well in processing large homogeneous objects such as ground and wall surfaces, thus lifting some workload for manual classification. Manual classification by polygons is more precise and accurate for the majority of areas, which have higher noise and occlusion.

4.2 Data Density and Coverage

Surveying at 15 stations ensured sufficient point density for the Block W Square and the TU plaza area. Moreover, the additional scanning locations (S14 and S15) helped to obtain additional points close to the benches. Despite the additional scans, certain areas obstructed by pillars and water-filled barriers still had lower point density. Multiple pre-processing methods were used, such as down sampling, denoising, and edge trimming. Moreover, edge trimming helped to filter out some raw data and to focus on analyzing the best part of the obtained points.

4.3 Challenges and Possible Solutions

The problem of occlusions related to pillars and water-filled barriers led to changing the location of some scanning stations. Moreover, the void filling function in 3DMas filled minor holes in the data under the tripod and other objects affected by occlusion. Automated segmentation also helped identify pedestrians who passed by during the scan of the TU plaza.

5. Conclusion

This project successfully completed the terrestrial laser scanning survey of Block W Square and adjacent TU Plaza using the Leica BLK360 G2. A detailed, geo-referenced point cloud has been collected and semantically segmented into ten distinct urban classes.

The practical application of this project has given firsthand experience on how to plan and conduct a terrestrial laser scanning survey. It also involved using software tools for pre-processing and classification of points. Hybrid segmentation, combining automated and manual tools, proved to be efficient in producing good quality segmented objects. This project showed that TLS can be effectively used to scan complex urban areas. Also, with proper segmentation, useful urban classes can be generated.

6. Contributions List

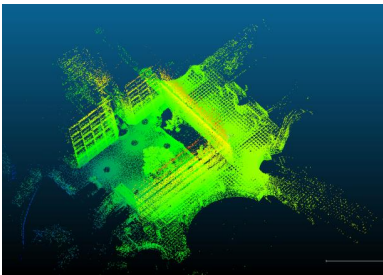
	████████ ██████	██████ ████████	██████ ██████	████████ ██████	████████ ██████
Leaders					
Field Survey					
Data Pre- processing					
Auto segmentation					
Void filling					
Semantic segmentation					
Digitized 2D Map					
Group Report					

7. Appendix

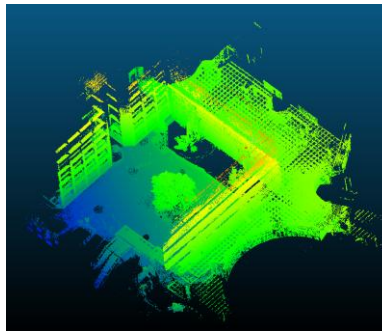
GNSS Survey Data

PointID	Northing	Easting	Height
201	818480.302	836641.219	12.134
1	818461.226	836643.897	12.111
2	818464.708	836644.802	12.348
3	818464.756	836645.101	12.146
5	818478.615	836660.114	12.222
6	818478.928	836660.202	12.182
201	818480.312	836641.217	12.133

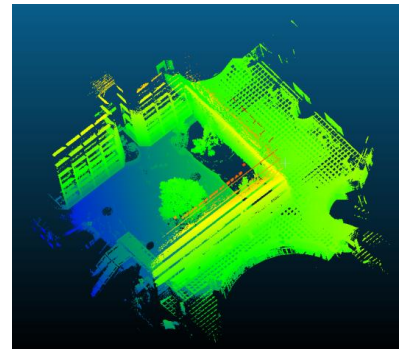
Before CloudCompare



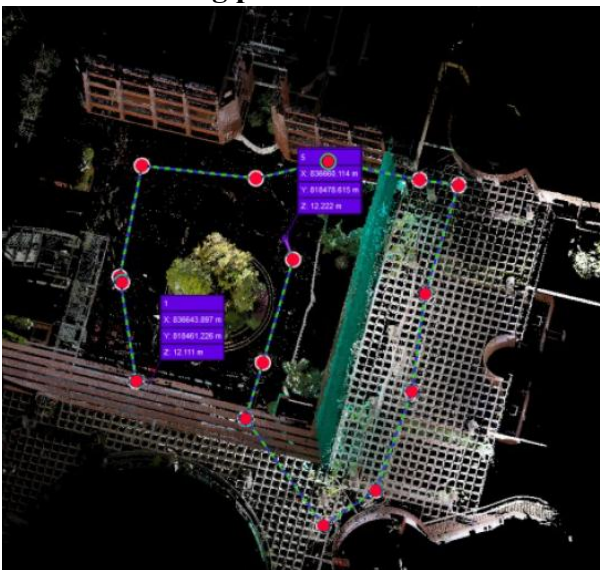
CloudCompare First Process



CloudCompare Final



Actual Scanning path



Link Quality Matrix

Link-Quality Matrix #1 -

